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1.

In the context of pathfinding, after reaching the target, a "reconstruct path" function is crucial for tracing the steps back to the starting point. This function relies on information collected during the search process about how each cell was reached. How does the "reconstruct path" function conceptually operate to accurately reconstruct the path from the target back to the starting point?

1 / 1 point

☐

By following a straight line from the target to the start based on their coordinates

☐

By decrementing a global path counter to reverse the sequence of moves

☒

By utilizing stored predecessors for each cell to trace the path in reverse from the target back to the starting point

☐

By randomly selecting adjacent cells to backtrack until the starting point is reached

☐

By scanning the entire grid again to find the shortest return path

☒

Correct

Correct. The function operates by utilizing information stored during the search about the predecessor of each cell (i.e., which cell we came from to reach this cell). Starting from the target, the function traces back through these predecessors, step by step, until it reaches the starting point. This method ensures that the reconstructed path accurately reflects the route taken by the search algorithm.

2.

The A* (A-star) algorithm is a highly efficient pathfinding and graph traversal method that combines the best features of Dijkstra's algorithm and the Greedy Best-First-Search to find the shortest path between a start and a target node in a weighted graph. At its core, what fundamental principle does the A* algorithm employ to ensure it efficiently finds the shortest path?

1 / 1 point

☐

It explores all possible paths from the start node to the target node before deciding on the shortest path

☐

It relies solely on the actual cost from the start node to a given node to determine the path

☒

It uses a heuristic to estimate the cost from a given node to the target node, combined with the known cost from the start node to the given node

☐

It randomly selects nodes to explore based on a probability distribution to ensure coverage of the graph

☐

It prioritizes nodes by their distance from the start node, ignoring any potential obstacles or weights

☒

Correct

Correct. The A* algorithm combines the actual cost from the start node to a given node (g cost) with a heuristic estimate of the cost from that given node to the target node (h cost). This estimated total cost (f cost = g cost + h cost) is used to prioritize the exploration of paths, focusing on those that appear most promising. This approach ensures an efficient search for the shortest path.

3.

In the A* (A-star) algorithm, the heuristic function is crucial for efficiently finding the shortest path between two points. The algorithm uses three key components: f, g, and h, in its decision-making process. What do these components represent within the context of the A* algorithm's heuristic function?

1 / 1 point

☒

f represents the total estimated cost of the cheapest path from the start node to the target through the current node, g represents the cost from the start node to the current node, and h represents the estimated cost from the current node to the target

☐

f represents the fixed cost of moving between any two adjacent nodes, g represents the gravitational pull affecting the path, and h represents the height difference between the current node and the target

☐

f represents the frequency of node visits, g represents the grid's geometry, and h represents the heuristic function's complexity

☐

f is the function used to filter out nodes with high costs, g stands for 'global positioning', and h signifies 'heuristic complexity'

☐

f is the formula used to calculate the algorithm's efficiency, g is the gauge of the algorithm's speed, and h measures the heuristic's accuracy

☒

Correct

Correct. This explanation accurately defines the roles of f, g, and h in the A* algorithm. f is the sum of g and h, providing an estimate of the lowest total cost path to the target that goes through the current node. g is the known cost from the start node to the current node, reflecting the path taken so far. h is the heuristic estimate of the cost to reach the target from the current node, guiding the algorithm's forward search.

https://www.coursera.org/learn/creative-coding-for-designers-using-python/assignment-submission/UdQpH/practice-quiz-pathfinding/view-feedback

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